UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF M.Sci. WITH HONOURS

FOURTH-YEAR FINAL EXAMINATION

03 23560

LM INFERENCE FROM SCIENTIFIC DATA

SUMMER EXAMINATION 2024

Time Allowed: 1 hour 30 minutes

Answer two questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

*Answer **two** questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. A continuous random variable $0 \leq x \leq 1$ follows a *beta distribution* if the probability density function (PDF) is of the following form,

$$f(x;\alpha,\beta) \propto x^{\alpha-1}(1-x)^{\beta-1}.$$

Here, $\alpha > 0$ and $\beta > 0$ are known as the *shape* parameters of the distribution.

(a) Normalise the expression for the PDF of the beta distribution. Your answer should consist of a complete, properly normalised expression for $f(x;\alpha,\beta)$; you may use the following result without proof: **[1]**

$$\int_0^1 \mathrm{d}t\; t^{z_1-1}(1-t)^{z_2-1} = \frac{\Gamma(z_1)\Gamma(z_2)}{\Gamma(z_1+z_2)} \quad \text{where } z_1, z_2 > 0.$$

(b) Find the mean of the beta distribution, giving your answer in terms of α and β. You may use without proof the fact that $\Gamma(z+1) = z\Gamma(z)$. **[2]**

(c) Find the variance of the beta distribution, giving your answer as a single fraction (over a common denominator) in terms of α and β. **[4]**

A biased coin has a probability $0 < \gamma < 1$ of giving a heads when it is flipped. The coin is flipped $N = 100$ times and $n = 33$ heads are obtained.

(d) Write down an expression for the likelihood $P(n|\gamma)$. **[2]**

In the remainder of the question you should use a beta distribution with $\alpha = \beta = 2$ as a prior on the parameter γ.

(e) Using Bayes' theorem, obtain an expression for the posterior $P(\gamma|n)$. You should make sure your posterior distribution is correctly normalised. **[10]**

(f) Using your answers from parts (b) and (c), find the mean and variance of the posterior $P(\gamma|n)$ from part (e). **[4]**

(g) Sketch the posterior distribution $P(\gamma|n)$ from part (e). **[3]**

Now the coin is to be flipped many more times to obtain an improved measurement of γ.

(h) Estimate the total number times the coin must be flipped in order to measure the parameter γ to a fraction uncertainty of 0.01%. You need only give your answer to the nearest order of magnitude. **[4]**

2. The value of a particular quantity X is measured N times and the following measurement values obtained:

$$\{x_i\} = \{x_1, x_2, \ldots, x_N\}\,.$$

The errors on each measurement are assumed to be independent. The error on the i^{th} measurement is assumed to be distributed as a Gaussian random variable with a known standard deviation of σ_i (the standard deviation is different for each measurement).

(a) Write down an expression for the likelihood $P(\{x_i\}|X)$. **[2]**

(b) Find the maximum likelihood estimator for the quantity X. **[4]**

For the remainder of this question you should use a Gaussian prior $P(X)$ (also know as a normal prior) with a mean value of μ and a standard deviation of Δ.

(c) Briefly explain what is meant by a *conjugate prior*. **[3]**

(d) Write down Bayes' theorem for this problem and obtain an expression for the posterior $P(X|\{x_i\})$. Show that the posterior is also a Gaussian and find the mean and variance of this distribution. **[12]**

(e) Find the mean and variance of the Gaussian posterior distribution $P(X|\{x_i\})$ found in part (d). **[7]**

(f) Find most likely value of X, i.e. the maximum posterior value of the parameter X. Compare this with the maximum likelihood estimator for X found in part (b) and comment on how these compare in the limit where an uninformative prior is used, $\Delta \to \infty$. **[2]**

3. A continuous random variable $x > 0$ follows a *gamma distribution* if the probability density function (PDF) is of the following form,

$$f(x; \alpha, \beta) \propto x^{\alpha-1} \exp(-\beta x).$$

Here, $\alpha > 0$ and $\beta > 0$ are known as the *shape* and *rate* parameters of the distribution respectively.

(a) Normalise the expression for the PDF of the gamma distribution. Your answer should consist of a complete, properly normalised expression for $f(x; \alpha, \beta)$. You may use without proof the following integral definition of the gamma function, $\Gamma(z)$: **[3]**

$$\Gamma(z) = \int_0^\infty \mathrm{d}t \; t^{z-1} \exp(-t) \quad \text{where } z > 0.$$

(b) Define the (non-central) moments μ_k of a continuous probability distribution. Also define the moment generating function (MGF) $M_x(t)$ of a continuous probability distribution. Explain how the MGF can be used to find the moments; you should include relevant formulae in your explanation. **[5]**

(c) Find the MGF $M_x(t)$ for the gamma distribution. **[5]**

(d) Using your result from part (c) or otherwise, find the mean and variance of the gamma distribution in terms of the parameters α and β. **[4]**

In an observation time T, an integer number $n \geq 0$ of events are observed to occur. Events are assumed to be independent and to occur at a constant average rate $\tilde{\lambda}$. (The parameter $\tilde{\lambda}$ has dimensions of time^{-1}.)

(e) Write down an expression for the likelihood of observing n events, $P(n|\tilde{\lambda})$. **[1]**

For the remainder of the question you should use the gamma distribution $f(x; \alpha, \beta)$ as a prior distribution for the rate parameter $\tilde{\lambda}$ with $x = \tilde{\lambda} T$.

(f) Using Bayes' theorem, find the posterior distribution for the rate parameter, $P(\tilde{\lambda}|n)$ and the Bayesian evidence, $P(n)$. You should express your answers for both $P(\tilde{\lambda}|n)$ and $P(n)$ in terms of α, β, n, and $\tilde{\lambda}$ (you may also use the gamma function) and you should give a correctly normalised expression for the posterior. **[12]**

Formula Sheet

Binomial Distribution.

$$P(r;n,p) = p^r(1-p)^{n-r}\,{}^nC_r$$

Poisson Distribution.

$$P(n;\lambda) = \frac{\lambda^n \exp(-\lambda)}{n!}$$

Normal Distribution. Also known as the Gaussian distribution.

$$P(x;\mu,\sigma^2) = \frac{\exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}}$$

Bayes' Theorem

$$P(A|B) = P(B|A)\frac{P(A)}{P(B)}$$

Physical Constants and Units

Acceleration due to gravity	g	9.81 $\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23}\ \mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\ \mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\ \mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\ \mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\ \mathrm{m^{-1}}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34}\ \mathrm{J\,s} \equiv 4.136 \times 10^{-15}\ \mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\ \mathrm{J\,s} \equiv 6.582 \times 10^{-16}\ \mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\ \mathrm{m\,s^{-1}}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	1.602×10^{-19} C
Mass of electron	m_e	9.109×10^{-31} kg
Rest energy of electron		0.511 MeV
Mass of proton	m_p	1.673×10^{-27} kg
Rest energy of proton		938.3 MeV
One atomic mass unit	u	1.66×10^{-27} kg
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12}\ \mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\ \mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\ \mathrm{A\,m^2}\ (\mathrm{J\,T^{-1}})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\ \mathrm{A\,m^2}\ (\mathrm{J\,T^{-1}})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	2.426×10^{-12} m
Bohr radius	a_0	5.2918×10^{-11} m
angstrom	Å	10^{-10} m
barn	b	$10^{-28}\ \mathrm{m^2}$
torr (mm Hg at 0 °C)	torr	133.32 Pa ($\mathrm{N\,m^{-2}}$)

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.